

Relieve or Aggravate? Impact of Interpersonal and Place Attachment Security Priming on Intrusive Symptoms

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Attachment security has the potential to be a protective factor against intrusive symptoms. However, its impact on intrusive symptoms across different types of trauma is not well understood. To address this, we explored how priming interpersonal and place attachment security affects intrusive symptoms in the context of man-made and natural traumatic events. One hundred sixty-five adult participants were randomly assigned to interpersonal or place attachment security priming or a control condition and subsequently watched man-made war or natural disaster films in a lab setting. For the following 7 days, they completed an intrusion diary each day. The results showed that although neither type of attachment security priming immediately alleviated distress following traumatic stimuli, both exerted mitigating effects during the subsequent week: Interpersonal attachment security reduced daily intrusion counts, and place attachment security decreased both intrusion counts and vividness. However, the interaction between man-made trauma and place attachment security priming was associated with worsened daily intrusion-related distress. Under natural disaster conditions, postpriming state attachment security and posttrauma reappraisal mediated the effects of attachment priming on intrusive and traumatic symptoms. This study reveals the varying effects of interpersonal or place attachment priming on intrusive memories in man-made war and natural disaster, demonstrating the impact of attachment security priming in the immediate and short term after viewing trauma films. These findings offer the potential for developing attachment-based interventions tailored to specific trauma types.

Keywords: attachment security priming, trauma film paradigm, intrusive symptoms, posttraumatic stress symptoms

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Traumatic events, whether man-made (e.g., war or violence) or natural (e.g., earthquakes or tsunamis), are associated with a range of adverse outcomes and frequently result in posttraumatic stress disorder (PTSD; Ye et al., 2023). PTSD is characterized by four primary symptom clusters: intrusive symptoms, avoidance, alterations in arousal and reactivity, and negative alterations in cognitions and mood (American Psychiatric Association, *DSM-5 Task Force*, 2013). Among these, intrusive symptoms are considered hallmark features and severely impair victims' daily functioning (James et al., 2016; Marks et al., 2018). Intrusive symptoms involve recurrent, involuntary, and distressing memories (e.g., images or thoughts) of traumatic events that intrude into consciousness, resulting in flashbacks, re-experiencing, and related phenomena (American

Psychiatric Association, *DSM-5 Task Force*, 2013; James et al., 2016). These symptoms are particularly detrimental to veterans (e.g., Solomon et al., 2009; Zakin et al., 2003) and survivors of natural disasters (e.g., Cheng et al., 2020), yet they are difficult to alleviate. To address this challenge, researchers have increasingly focused on the role of attachment security in intrusive symptoms (e.g., Bryant & Datta, 2019; Karl et al., 2021).

Attachment security can manifest in interpersonal or place contexts. Interpersonal attachment security refers to individuals consistently perceiving their attachment figures as available, sensitive, and attentive to their needs, while maintaining a positive internal working model in which they view themselves as lovable and others as trustworthy and reliable (Ainsworth et al., 1978).

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According to attachment theory (Bowlby, 1961, 1969, 1973), attachment security is crucial for coping with threats and trauma by fostering a sense of safety and regulating distress. This perspective is further supported by the framework of attachment processes in PTSD (Bryant, 2023) and the social-cognitive model of PTSD (Sharp et al., 2012). The social-cognitive model of PTSD (Sharp et al., 2012) posits that, when facing trauma, individuals with attachment security activate attachment-related schemas, which facilitate adaptive social-cognitive processing and promote effective use of existing attachment relationships or social support systems to buffer against the negative effects of trauma. Meta-analytic evidence demonstrates a significant association between attachment security and lower PTSD symptom severity (Cushing et al., 2024; Woodhouse et al., 2015). Intervention studies further show that enhancing attachment security predicts reductions in overall symptom severity and traumatic symptoms (e.g., intrusive symptoms) after treatment (Muller & Rosenkranz, 2009). Moreover, experimental evidence suggests that the activation of interpersonal attachment security can mitigate intrusive memories (e.g., Bryant & Datta, 2019; Karl et al., 2021).

Correspondingly, place attachment, a key construct in environmental psychology, has garnered increasing attention as a potential protective factor for mental functioning (Scannell & Gifford, 2017b). Place attachment security refers to the emotional and cognitive bonds that connect individuals to specific locations (Bonaiuto et al., 2016). The tripartite framework of place attachment (Scannell & Gifford, 2010) emphasizes its multifaceted functions, including providing survival and security, supporting self-regulation and goal pursuit, ensuring continuity, and fostering belonging, thereby offering protection against threats (Giuliani, 2003). However, no studies to date have directly examined the influence of place attachment on intrusive symptoms. Beyond this, prior studies on the association between attachment security and intrusive symptoms have some limitations that remain unaddressed. For instance, these studies often relied on retrospective reports, which obscured peri- and immediate posttraumatic processes and may introduce reporting bias. Moreover, the absence of control groups in previous studies raises questions about whether intrusive symptoms can be directly attributed to traumatic events. Furthermore, the mechanisms underlying attachment security that may influence intrusive symptoms were not well characterized by previous cross-sectional studies.

To address these issues, we employed attachment security priming to activate interpersonal and place attachment security in a laboratory setting. This paradigm has been widely used to elicit mental representations of secure attachment figures (Mikulincer et al., 2001) or places (Scannell & Gifford, 2017a), temporarily increasing state security (Gillath & Karantzas, 2019). Meanwhile, the trauma film paradigm was used widely in trauma psychology to study peri- and immediate posttraumatic processes (James et al., 2016). By exposing nonclinical participants to analogue traumatic events, this method reliably elicits short-term intrusive symptoms similar to those following real trauma, which typically subside within a week (Holmes et al., 2004; James et al., 2016; Varma et al., 2024). Accordingly, a 1-week follow-up window was adopted in the original design (Holmes et al., 2004, 2009), and this time frame is widely used to capture the formation, consolidation, and retrieval of intrusive memories (James et al., 2016; Karl et al., 2021), as well as delayed effects that are not observable through immediate

assessment. Therefore, the present study employed the trauma film paradigm with daily assessments over 1 week to examine the effects of interpersonal and place attachment security on intrusive symptoms and their underlying mechanisms.

The Impact of Interpersonal and Place Attachment Security on Intrusive Symptoms After Man-Made and Natural Trauma

Both interpersonal and place attachment provide a secure base, protecting individuals from threats and offering psychological benefits (Cushing et al., 2024; Scannell & Gifford, 2017b), and they share key psychological processes (Scannell et al., 2020). However, no research has identified significant differences between their roles in relation to traumatic symptoms. Regarding their origins, interpersonal attachment is grounded in stable and secure relationships with significant others, and its benefits are well-documented (Cassidy & Shaver, 2016). In contrast, place attachment extends to nonhuman entities (e.g., places and external environments) and develops across time and space, so the stability of its effects warrants further investigation (Scannell et al., 2020).

These distinct characteristics of interpersonal and place attachment security under different types of trauma may result in varying effects on traumatic symptoms. Among them, trauma can be categorized by its causes as either natural (e.g., earthquakes) or man-made (e.g., wars; Zagefka et al., 2010). Both types are associated with a high prevalence of PTSD but differ in core symptomatology. For example, earthquakes, marked by suddenness and uncontrollability, are associated with high-impact symptoms such as detachment, estrangement, and nightmares (Qi et al., 2023). Meanwhile, wars involve blame and anger directed at specific targets, with survivors exhibiting intrusive symptoms and avoidance as central features in the maintenance of PTSD (DiMauro et al., 2014; Schlechter et al., 2022). Given that trauma types may show distinct mechanisms, a critical research gap concerning how interpersonal and place attachment security affect intrusive symptoms across these types warrants attention and empirical investigation.

In fact, the relationship between interpersonal attachment and traumatic symptoms is influenced by the type of trauma (Barazzzone et al., 2019). Secure interpersonal attachment can reduce PTSD symptoms following natural disasters (e.g., Tian et al., 2020); however, in the context of man-made trauma, particularly war, attachment security may be associated with increased PTSD symptoms. Specifically, individuals with high interpersonal attachment security may experience powerlessness and loss of control during natural disasters (Zhou et al., 2018). Such experiences may evoke negative awe toward the environment and motivate these individuals to actively establish social connections with others (Xu, 2018). This process fosters trust and reliance on others, reinforces positive attitudes toward social support (Cassidy & Shaver, 2016), and thus alleviates intrusive PTSD symptoms. When confronted with man-made trauma, individuals with high levels of interpersonal attachment security tend to attribute the trauma to human agents (Zagefka et al., 2010). This attribution can challenge positive internal representations of others, thereby disrupting a sense of security and basic trust in others (Barazzzone et al., 2019; Charuvastra & Cloitre, 2008). As a result, individuals' capacity to maintain secure attachments may be

undermined, leading to increased tendencies toward defensive interpersonal avoidance or heightened attachment anxiety (Solomon et al., 2008). Such processes may contribute to more intrusive symptoms than those observed following natural disasters.

However, the influence of place attachment on intrusive symptoms following both man-made and natural traumas remains unexplored. Actually, person–place bonds are not invariably positive and can sometimes be negative, ambivalent, or even dangerous (Scannell et al., 2020), particularly after natural disasters. When individuals experience natural disasters, attachment to places may provide a “false” sense of security, leading to reluctance to relocate due to concerns about the destruction and loss of meaningful and familiar places (Bonaiuto et al., 2016; Silver & Grek-Martin, 2015). It may also result in disruptions to their sense of identity and collective dislocation from community and belonging (Cox & Perry, 2011). These disruptions may intensify individuals’ focus on the negative aspects of the trauma, heightening perceptions of danger and threat, and subsequently increasing the risk of PTSD symptoms (Lonigan et al., 1994; Puechlong et al., 2021), including intrusive symptoms. Moreover, flood victims with stronger place attachment to their home and city have been found to exhibit higher levels of PTSD symptoms (Puechlong et al., 2021). Similarly, individuals who lost their homes in natural disasters reported more severe PTSD symptoms (Lonigan et al., 1994) and heightened stress (Felix et al., 2015). However, for individuals who have experienced man-made trauma, place attachment might provide a safe haven and act as a temporary escape from chaos (Bonaiuto et al., 2016), increase self-esteem, foster a sense of belonging, and provide meaning (Scannell & Gifford, 2017a), thereby protecting individuals from threats (Giuliani, 2003) and potentially reducing intrusive symptoms.

Based on this, we hypothesize a significant interaction between attachment priming and trauma type on intrusive symptoms (Hypothesis 1). Specifically, primed interpersonal attachment security is expected to reduce intrusive symptoms following natural disasters but increase intrusive symptoms following man-made trauma. Conversely, primed place attachment security is expected to increase intrusive symptoms after natural disasters but decrease intrusive symptoms after man-made trauma.

Mechanisms Underlying the Impact of Attachment Security on Intrusive Symptoms

According to attachment theory (Bowlby, 1980), securely attached individuals are more likely to manage posttraumatic responses by activating state attachment security and employing adaptive emotion regulation strategies. Under experimental conditions, attachment security priming activates mental representations of attachment figures in memory, which may provide protection, comfort, and support, thereby enhancing state attachment security and the sense of safety (Gillath et al., 2022). When individuals experience secure relational bonds, they are more open to integrating new information and developing novel understandings (Gross, 2014), facilitating the reappraisal of traumatic events (Zhou et al., 2018). Empirical evidence indicates that attachment security is positively associated with increased use of reappraisal (Karreman & Vingerhoets, 2012; Tammilehto et al., 2022; Troyer & Greitemeyer, 2018). In the process model of emotion regulation (Gross & Thompson, 2007), the use of reappraisal strategies can prompt individuals to alter their cognitive

interpretation or evaluation of the situation (Gross, 1998). In the long term, this process helps individuals reconstruct the meaning of traumatic experiences, alleviate distress, and achieve posttraumatic adaptation (Zhou et al., 2016). Reappraisal has been shown to reduce the severity of PTSD, particularly intrusive memories (Woud et al., 2012; Zhou et al., 2018), and its beneficial effects on PTSD tend to be more pronounced over time following trauma (Nowlan et al., 2016; Sears et al., 2003).

Numerous studies have found that the most direct impact of attachment security priming is an increase in feelings of security and state attachment security (Gillath et al., 2022). Feeling secure in attachment enables individuals to maintain stable self-worth and emotional equanimity (Mikulincer & Shaver, 2020), thus reducing the impact of traumatic events and inhibiting the intrusion of distressing memories (Bryant & Chan, 2017). In addition, attachment security priming facilitates emotion regulation and may alter the interpretation of traumatic events to be more positive or less threatening, thereby enhancing reappraisal (Gross, 2014; Mikulincer & Shaver, 2020). Reappraisal has the potential to expand one’s perspectives and mental flexibility while also alleviating distress (Tammilehto et al., 2022), thereby reduces the severity of intrusive symptoms of PTSD in the long term. While most evidence supports the long-term efficacy of reappraisal in reducing PTSD, the immediate effects of posttrauma reappraisal have yet to be examined.

Based on the above, we hypothesize that attachment priming reduces intrusive memories following trauma through increasing postpriming state attachment security and enhancing posttrauma reappraisal (Hypothesis 2).

The Present Study

While previous studies have examined the association between attachment security and intrusive symptoms, several limitations remain to be addressed. First, prior studies primarily focused on the impact of interpersonal attachment security on intrusive symptoms (e.g., Bryant & Datta, 2019), but they overlooked the role of place attachment security. Second, previous studies using trauma film paradigms often mixed different types of traumatic films together for viewing (e.g., Holmes et al., 2004; Karl et al., 2021). As a result, the findings cannot provide specific insights into a particular type of trauma, since each has its own unique characteristics. Thus, it remains unclear how the effects of attachment security on intrusive symptoms differ across various attachment figures and types of traumatic events, which is important for delineating their boundary conditions. Third, according to models of intrusive memories (Brewin et al., 2010) and prior research (Krinke et al., 2022; Rombold et al., 2016; Varma et al., 2024), intrusive symptoms encompass not only frequency but also vividness (sensory-perceptual qualities) and distress (emotional intensity). However, some studies have focused solely on intrusion frequency and have not directly examined the effects of attachment security on the different aspects of intrusive symptoms. Fourth, most prior studies have relied on self-reported measures, which are susceptible to recall bias. In addition, few studies have used daily experience sampling to measure daily changes in intrusive symptoms; thus, little attention has been given to the immediate and short-term effects of attachment security on daily intrusive symptoms. Finally, the potential psychological mechanisms underlying these effects have not been thoroughly examined.

To address the aforementioned gaps, this study aims to (a) examine potential differences in the effects of attachment figures and trauma types on intrusive symptoms and (b) investigate whether postpriming state attachment security and posttrauma reappraisal mediate the relationship between attachment security and intrusive symptoms.

Method

Participants

Previous studies recruited 25–35 participants per condition and found a medium to large effect size ($f = 0.30$; e.g., Bryant & Chan, 2017; Karl et al., 2021). Therefore, based on the effect sizes reported in prior studies, we conducted a power analysis using G*Power 3.0 (Faul et al., 2007). The analysis indicated that a sample size of at least 25 participants per condition is required to achieve 80% power for analyses of variance (ANOVAs) at an α level of .05. Accordingly, we recruited 179 adult participants through posters or public elective courses at a university campus in Zhejiang University. They were randomly assigned to one of six groups in a 3 (attachment security priming: interpersonal, place, control) $\times$ 2 (trauma: man-made war, natural disaster) between-subjects design (see Supplemental Figure S1). Referring to previous studies (e.g., Holmes et al., 2009; Xu et al., 2024), we excluded participants who met any of the following criteria: (a) a history of heart disease, epilepsy, schizophrenia, or PTSD, which was emphasized during recruitment; (b) a diagnosis of depression, anxiety, or other mental health disorders ($n = 6$); (c) failure to follow the writing instructions as assessed by two experimenters ($n = 3$); or (d) incomplete participation due to missing some experimental steps ($n = 5$). Following the above exclusions, we included 165 participants in the main analysis: 70 males (42.4%) and 95 females (57.6%) with an average age of 19.92 years ($SD = 1.99$). Two participants were further excluded for not completing the intrusion diary within 7 days, so 163 participants were included in the analysis of daily data: 69 males (42.3%) and 94 females (57.7%) with an average age of 19.94 years ($SD = 2.00$ years). This study was approved by the Research Ethics Committee of the Department of Psychology and Behavioral Sciences at Zhejiang University. In line with the ethics guidelines, participants were free to withdraw from the survey at any time.

Measures

Trait Attachment

To assess the insecure attachment traits in close relationships, we utilized the Experiences in Close Relationships Inventory developed by Brennan et al. (1998), which included both attachment anxiety and avoidance subscales. The subscales each contained 18 items rated on a 7-point Likert scale, ranging from 1 (*completely disagree*) to 7 (*completely agree*). The Cronbach's α coefficient for both subscales were .89, which showed good reliability in our study.

Place Attachment

The Place Attachment Scale (Williams & Vaske, 2003) was used to assess individuals' attachment to places where they have lived for a long time. The scale comprised 12 items that were rated on a 5-point

Likert scale, ranging from 1 (*strongly disagree*) to 5 (*strongly agree*). A higher score indicated a stronger attachment to the place. It showed good reliability in this study ($\alpha = .86$).

Depression

The Patient Health Questionnaire–9 (Kroenke et al., 2001) was used to assess symptoms of depression over the past 2 weeks. It consisted of nine items rated on a 4-point Likert scale (0 = *not at all*, 3 = *nearly every day*) and demonstrated good reliability at baseline ($\alpha = .80$).

Security

The Felt-Security Scale (Luke et al., 2012) was used to measure state security with 16 items rated on a 6-point Likert scale, ranging from 1 (*not at all*) to 6 (*very much*). A higher score on the scale indicated a greater sense of security. The scale used in this study showed strong reliability, with α of .95 at Time 0 (T0) and .96 at both Time 1 (T1) and Time 2 (T2).

State Attachment

We used a simplified version of the State Adult Attachment Measure (Gillath et al., 2009), revised by Tammilehto et al. (2022), to measure state attachment at the moment. This scale consisted of six items, two of which were specifically designed to assess state attachment security. These subscales used a 7-point Likert scale (1 = *strongly disagree* to 7 = *strongly agree*), and the Cronbach's α coefficients were .88–.94 across the T0–T2 phases and showed good reliability in previous studies (e.g., Tammilehto et al., 2022).

Positive and Negative Affect

The Positive and Negative Affect Schedule included two subscales, positive affect and negative affect, each containing 10 emotional descriptors (Watson et al., 1988). On a 5-point Likert scale (1 = *very slight or not at all* to 5 = *extremely strong*), participants were asked to rate the intensity of the affect they experienced. In this study, both positive affect (with α of .93 at T1 and .86 at T2) and negative affect (with α of .73 at T1 and .89 at T2) demonstrated good reliability.

Trauma Film Ratings

According to Holmes et al. (2004) and Karl et al. (2021), we used four items to measure aspects of trauma film processing on Likert scales. Participants were asked to rate the level of attention they paid to the films (0 = *not at all*, 10 = *total attention*), the degree of distress caused by the films (0 = *not at all*, 10 = *extremely*), the level of personal relevance of the films (0 = *not at all*, 10 = *extremely*), and the perspective from which they viewed the films (field vs. observer perspective; -3 = *entirely looking out through my own eyes*, $+3$ = *entirely observing myself from an external point of view*).

Reappraisal

We utilized the State Emotion Regulation Questionnaire (Brans et al., 2013), which consisted of six items, each describing a different emotion regulation strategy. Each item was rated on a 6-point Likert

scale (0 = *not at all*, 5 = *very much so*). We only measured state reappraisal, which was described as “I have changed the way I think about what causes my feelings.” The validity and reliability of this scale were found to be satisfactory (Brans et al., 2013).

Intrusion Diary

We used the intrusion diary to assess participants' intrusion count, intrusion-related vividness, and intrusion-related distress related to the movie clips (Holmes et al., 2004). Participants were required to monitor and report the number of flashback memories related to the movie clips (the sum of images and thoughts) every day, as well as the vividness (0 = *completely blurry*, 10 = *very vivid*) and distress (0 = *not at all distress*, 10 = *extremely distress*) of their memories on that day. This diary has been used in several previous studies (e.g., Holmes et al., 2004) and demonstrated low to moderate within-group variability in intrusion count (intraclass correlation coefficient = .30), vividness (intraclass correlation coefficient = .38), and distress (intraclass correlation coefficient = .53).

Posttraumatic Stress Symptoms

The Impact of Event Scale–Revised (IES-R; Weiss, 2007) was used to measure posttraumatic stress symptoms (PTSSs) and intrusion related to trauma films 1 week after the experiment. This scale has been widely applied in trauma film paradigm research (e.g., Badawi et al., 2020; Varma & Hu, 2022). The IES-R consisted of 22 items including three dimensions: intrusion (e.g., “pictures about it popped into my mind”), avoidance (e.g., “stayed away from reminders of it”), and hyperarousal (e.g., “I had trouble concentrating”), rated on a 5-point Likert scale ranging from 0 (*not at all*) to 4 (*extremely*). The total PTSSs score was calculated by summing all item scores. The IES-R and its intrusion subscale demonstrated good reliability ($\alpha = .93$ and $\alpha = .89$, respectively).

Experimental Tasks and Stimulus Material

Priming

To prime participants' interpersonal attachment security, we asked them to recall and write about an important safe relationship they had established with others (interpersonal attachment condition; Bartz & Lydon, 2004). When priming their attachment security to a place, we modified the instruction by asking participants to recall and write about a familiar place to which they felt attached (place attachment condition; Scannell & Gifford, 2017a). As a neutral control condition, participants were asked to write down their schedules for the previous day (adapted from Baikie et al., 2012). To ensure adequate priming, we instructed participants in each group to handwrite for at least 8 min (e.g., Karl et al., 2021). Instructions for priming and examples of writing samples can be found in Supplemental Materials.

Trauma Film Paradigm

This paradigm has been used in numerous previous studies (with 4–12 min film clips; Badawi et al., 2020; Bryant & Datta, 2019; Holmes et al., 2009; Karl et al., 2021; Varma & Hu, 2022; Xu et al., 2024) to evoke brief trauma-analogue responses, such as intrusions and other PTSD-like symptoms, which typically subside within a

week (James et al., 2016; Varma et al., 2024). The trauma film paradigm has gained increasing recognition with the release of the *Diagnostic and Statistical Manual of Mental Disorders*, fifth edition, which notes that actual PTSD symptoms can arise from trauma exposure via media, television, movies, or pictures when such exposure is work-related (American Psychiatric Association, *DSM-5 Task Force*, 2013; Varma et al., 2024). Therefore, this study applied this paradigm with each film clip lasting approximately 6 min. The natural disaster condition utilized clips from the movie *AfterShock*, which depicts scenes of collapsed houses, casualties, and ruins after an earthquake happened in Tangshan. The man-made war condition used clips from the movie *City of Life and Death*, which mainly depicts scenes of Japanese soldiers killing soldiers and civilians in Nanjing using burning, stabbing, burying alive, shooting, and other means. Participants were asked to watch the film clips carefully.

Procedure

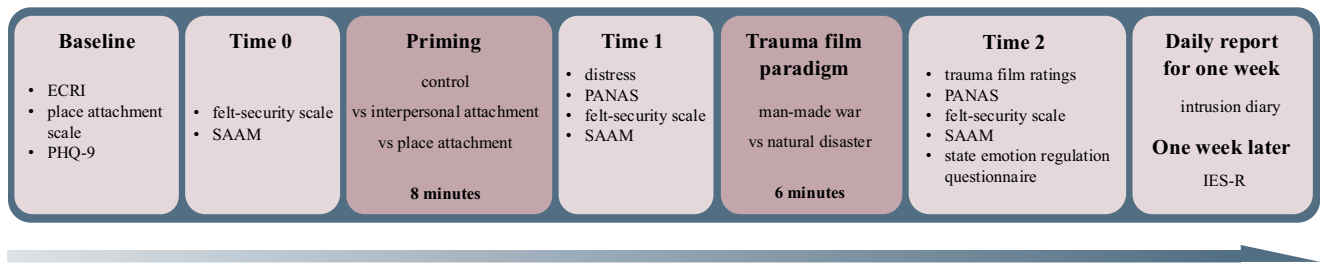
Upon arrival at the laboratory, participants signed a written informed consent form and completed a baseline questionnaire to collect their gender, age, trait attachment, place attachment, and depression. Participants were randomly assigned to six experimental conditions, with the same experimental procedure for each condition as shown in Figure 1. Before the writing task (T0), participants were asked to watch a scenery video (a peaceful sea surface connecting with the sky, accompanied by gentle waves) to maintain a calm mood and completed a questionnaire to measure their security and state attachment security. Participants were then instructed to write for 8 min based on the instructions of interpersonal attachment, place attachment, or control condition. After completing the writing task (T1), they filled out a questionnaire regarding their security, state attachment security, positive and negative affect, and state distress. Next, using the trauma film paradigm, participants watched a traumatic film clip (depicting man-made wars or natural disasters) corresponding to their randomly assigned condition to induce a traumatic experience. The relevant variables were measured (T2; see Figure 1). Finally, participants ended the experiment by watching a landscape movie to regain a good mood.

After the experiment, a brief interview and assessment were conducted for each participant to ensure that they left the laboratory in a good state with stable emotions. After leaving the laboratory, participants were required to fill out an intrusion diary every day online for 7 days to assess the daily intrusion count, vividness, and distress related to the film clips. Participants who did not fill out the diary for more than 2 days were considered invalid and were notified by the experimenter to withdraw from the study. One week after the end of the experiment, participants were asked to complete the IES-R. Finally, participants who completed the experiment were compensated with either 60 min of experimental time as bonus points for the course or 30 RMB.

Statistical Analyses

Data analysis was performed using SPSS 26.0 and Mplus 8.3. ANOVA was used to assess randomization effectiveness and participant homogeneity based on gender, age, place attachment, trait attachment avoidance/anxiety, depression, and T0 variables across six groups. Furthermore, we used analyses of covariance and *t* tests

Figure 1
Experimental Procedure



Note. ECRI = Experiences in Close Relationships Inventory; IES-R = Impact of Event Scale–Revised; PHQ-9 = Patient Health Questionnaire–9; PANAS = Positive and Negative Affect Schedule; SAAM = State Adult Attachment Measure. See the online article for the color version of this figure.

to conduct manipulation checks, which examined the effects of attachment priming and the trauma film paradigm. Statistical significance was set at $\alpha = .05$. Depending on the stage of the study, data were categorized as experimental or daily and analyzed in three steps:

1. For experimental data analysis, we used ANOVAs to further examine the main effects and interactions of attachment priming and trauma on the T2 variables. Post hoc comparisons with Bonferroni multiple corrections were performed in cases of significant main or interaction effects.
2. For daily data analysis, descriptive statistics were computed, and line charts were plotted to visualize changes in daily intrusion count, intrusion-related vividness, and intrusion-related distress across groups. To test Hypothesis 1, we examined the effects of attachment priming condition (interpersonal attachment vs. place attachment vs. control), trauma type (man-made war vs. natural disaster), and their interaction on daily intrusion count, intrusion-related vividness, and intrusion-related distress using multilevel linear models with random intercepts, as well as on IES-R scores at 1-week follow-up using linear regression models. Multilevel models accounted for the nesting of repeated measures, such that daily intrusion count, intrusion-related vividness, and intrusion-related distress were treated as within-person (Level 1) variables, whereas attachment priming condition, trauma type, T1 state attachment security, and T2 reappraisal were treated as between-person (Level 2) variables. Dummy coding was applied, with the control condition as the reference group for attachment priming and the natural disaster condition as the reference for trauma type. For significant interaction effects, we further conducted simple effect analyses in Mplus 8.3 by comparing the estimated means of daily intrusion outcomes at the within-person level across attachment priming conditions within each trauma type condition.
3. To test Hypothesis 2 and to clarify the interaction between attachment and trauma conditions, separate models were constructed for the man-made war and natural disaster conditions. Under both trauma conditions, we assessed the mediating effects of T1 state attachment security and T2 reappraisal on the associations between attachment priming and daily intrusion count, intrusion-related

vividness, and intrusion-related distress (using 2–2–1 multilevel mediation analysis; Preacher et al., 2010), as well as on IES-R scores (using mediation models). In these analyses, the control condition was dummy coded as the reference group.

All analyses for Hypotheses 1 and 2 were conducted in Mplus 8.3 using Bayesian estimation with two independent Markov chain Monte Carlo chains. This approach treats parameters as random variables and combines prior information with observed data to obtain posterior distributions. It is well-suited for small samples and outperforms parametric bootstrap and Monte Carlo methods (Fang et al., 2019). In all multilevel linear model analyses, participant was included as a random factor (i.e., grouping variable) to account for individual-level variability. Results are reported with 95% credible intervals (CIs), and indirect effects were considered significant when the CIs did not include zero. Coefficients represent the relative contribution of each condition compared to the reference group. Missing data were handled automatically by the Markov chain Monte Carlo algorithm (da Silva Castanheira et al., 2024).

Transparency and Openness

The data are available from the corresponding author. The analysis code is openly available in the [Supplemental Materials](#). The design and analytical procedures of this study were not preregistered.

Results

Randomization Checks

ANOVA on gender, age, place attachment, trait attachment anxiety, trait attachment avoidance, depression at baseline, and state security and state attachment security at T0 across the six groups revealed no significant differences ($ps > .05$; see [Supplemental Table S1](#)). These results indicate that participants were relatively homogeneous, and the randomization to experimental conditions was successful.

Manipulation Checks

For attachment priming, within-group analyses using paired t tests showed that attachment security was effectively primed, as evidenced by increased security and state attachment security from

T0 to T1 in both the interpersonal and place attachment conditions (see Supplemental Table S2). For trauma film processing, the effectiveness of the trauma film paradigm was validated, as participants in both the man-made war and natural disaster conditions reported increased negative affect, decreased positive affect, and higher distress after viewing the trauma films compared to before. Moreover, participants rated the films as highly personally relevant, with no clear preference for either a field or observer perspective. Further details are provided in the Supplemental Materials.

Experimental Data Analysis

ANOVAs of the Between-Condition Differences in T2 Variables

Supplemental Table S3 presents ANOVA results for T2 variables under different conditions. For state attachment security, significant main effects were found for attachment priming and trauma, but not their interaction. Post hoc Bonferroni-corrected comparisons showed lower state attachment security in place attachment conditions compared to control, $t(107) = -2.50$, $p_{\text{Bonferroni}} = .041$, $d = -0.48$. Man-made war condition resulted in lower state attachment security than natural disaster condition, $t(163) = -3.12$, $p_{\text{Bonferroni}} = .002$, $d = -0.49$. More importantly, we plotted the changes in state attachment security at T0, T1, and T2 across attachment priming and trauma conditions (see Figure 2). We found that state attachment security in the control condition remained stable over time. In contrast, attachment priming led to an increase in state attachment security immediately following the priming. Unexpectedly, as shown in Figure 2 and confirmed by paired-sample t tests comparing T1 and T2, there was a significant decrease in state attachment security after viewing the trauma film in both the interpersonal attachment condition: $t(55) = -4.34$, $p_{\text{Bonferroni}} < .001$, $d = -0.16$, and the place attachment condition: $t(80) = -5.42$, $p_{\text{Bonferroni}} < .001$, $d = -0.60$. Moreover, a substantial decline was also observed in the man-made war condition: $t(80) = -5.42$, $p_{\text{Bonferroni}} < .001$,

$d = -0.60$. These findings suggest that attachment priming may increase vulnerability to subsequent trauma exposure.

For distress, security, and reappraisal at T2, no significant main or interaction effects of attachment priming and trauma were observed, indicating that attachment activation did not immediately alleviate distress but decreased state attachment security.

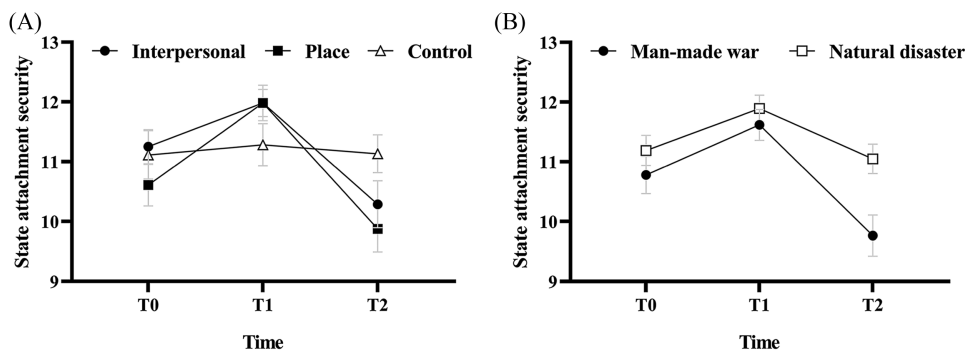
Daily Data Analysis

Hypothesis 1: Multilevel Models in Daily Intrusion Count, Intrusion-Related Vividness, Intrusion-Related Distress, and IES-R

A multilevel analysis was conducted with attachment priming condition, trauma condition, and their interaction as between-level independent variables, and daily intrusion count, vividness, and distress as within-level dependent variables (see Table 1). When analyzing daily intrusion counts, we found main effects of both interpersonal/place attachment priming and trauma type. Specifically, compared to the control condition, both interpersonal and place attachment priming predicted fewer daily intrusions. Exposure to man-made war films was associated with lower daily intrusion counts compared to exposure to natural disaster films. Moreover, there was a significant interaction between attachment priming and trauma type on daily intrusion counts. Further simple effects analyses (see Figure 3A) indicated that under the natural disaster condition, both interpersonal attachment priming (simple effect = -0.442 , $p = .004$, 95% CI $[-0.77, -0.11]$) and place attachment priming (simple effect = -0.343 , $p = .023$, 95% CI $[-0.68, -0.01]$) were associated with fewer intrusions compared to the control condition. Within the control condition, participants exposed to natural disaster films reported more intrusions than those exposed to man-made war films (simple effect = -0.36 , $p = .018$, 95% CI $[-0.70, -0.03]$).

For analyses of daily intrusion-related vividness, we found a main effect of place attachment priming, but not of trauma type, as well as a significant interaction between place attachment priming and

Figure 2
Changes in State Attachment Security Across Time by (A) Attachment Priming Condition and (B) Trauma Types



Note. (A) State attachment security at baseline (T0), after priming (T1), and after trauma induction (T2) for interpersonal attachment, place attachment, and control conditions. (B) State attachment security at baseline (T0), after priming (T1), and after trauma film paradigm (T2) for man-made war and natural disaster conditions. Interpersonal = interpersonal attachment security priming condition; Place = place attachment security priming condition; Control = control condition. T0 = Time 0; T1 = Time 1; T2 = Time 2.

Table 1
Multilevel Models for Daily Intrusion Count, Intrusion-Related Vividness, and Intrusion-Related Distress, and Regression Model for IES-R

Indicator	Daily intrusion count			Daily intrusion-related vividness			Daily intrusion-related distress			IES-R intrusion			IES-R		
	β	95% CI	<i>p</i>	β	95% CI	<i>p</i>	β	95% CI	<i>p</i>	β	95% CI	<i>p</i>	β	95% CI	<i>p</i>
Interpersonal attachment priming	-.37	[-0.62, -0.10]	.004	-.16	[-0.41, 0.10]	.119	-.07	[-0.31, 0.18]	.295	-.002	[-0.25, 0.24]	.493	-.06	[-0.30, 0.19]	.328
Place attachment priming	-.29	[-0.54, -0.01]	.023	-.28	[-0.53, -0.01]	.021	-.11	[-0.36, 0.14]	.193	.06	[-0.19, 0.30]	.330	.01	[-0.24, 0.26]	.464
Man-made war	-.32	[-0.59, -0.02]	.018	-.11	[-0.38, 0.18]	.232	-.07	[-0.33, 0.21]	.319	.09	[-0.18, 0.34]	.258	.04	[-0.23, 0.29]	.394
Interpersonal Attachment Priming × Man-Made War	.28	[-0.03, 0.56]	.036	.17	[-0.12, 0.45]	.126	.07	[-0.20, 0.34]	.306	.01	[-0.26, 0.28]	.469	.08	[-0.20, 0.34]	.291
Place Attachment Priming × Man-Made War	.37	[0.06, 0.65]	.009	.41	[0.12, 0.68]	.003	.34	[0.05, 0.60]	.011	-.14	[-0.40, 0.14]	.171	-.05	[-0.32, 0.22]	.353

Note. All regression coefficients presented are standardized coefficients. The control condition was used as the reference group for attachment priming, and the natural disaster condition was used as the reference group for trauma type. Values in bold indicates significant results. IES-R = Impact of Event Scale-Revised; CI = credible interval.

trauma type. Further simple effects analyses (see Figure 3B) revealed that under natural disaster conditions, participants in the place attachment priming group reported significantly lower daily intrusion-related vividness compared to the control condition (simple effect = -1.01 , $p = .021$, 95% CI $[-1.97, -0.04]$). Daily intrusion-related vividness in the place attachment priming group was significantly higher under the man-made war condition than under the natural disaster condition (simple effect = 1.54 , $p = .001$, 95% CI $[0.57, 2.50]$).

For analyses of daily intrusion-related distress, we found no main effects of place attachment priming or trauma type, but a significant interaction between place attachment priming and trauma type. Further simple effects analyses (see Figure 3C) indicated that under the man-made war condition, participants in the place attachment priming condition reported significantly greater daily intrusion-related distress than those in the control condition (simple effect = 1.05 , $p = .008$, 95% CI $[0.19, 1.90]$). Daily intrusion-related distress in the place attachment priming group was significantly higher under the man-made war condition than under the natural disaster condition (simple effect = 1.23 , $p = .003$, 95% CI $[0.36, 2.08]$). Besides, regression analyses using IES-R and IES-R intrusion scores as dependent variables revealed no significant predictive effects of attachment priming, trauma type, or their interaction.

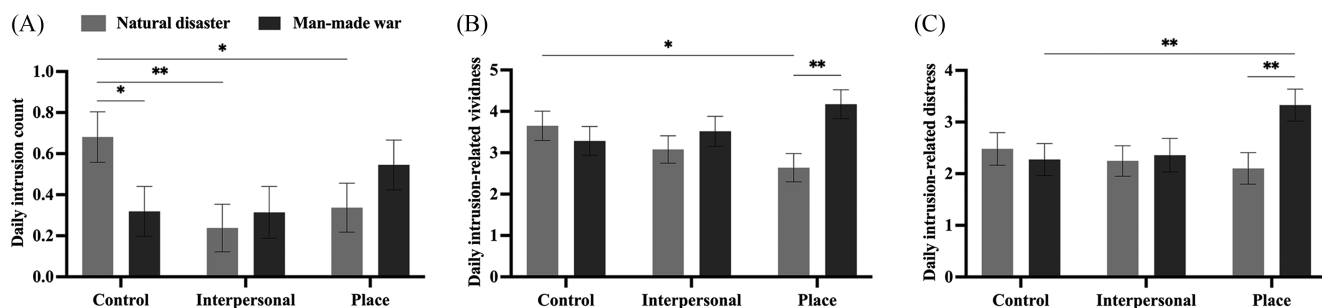
In addition, we examined the effects of attachment priming and trauma type on intrusive symptoms for each day and plotted the trajectories of intrusion count, vividness, and distress across each day within 1 week (Supplemental Figures S2–S4). These findings and further details are provided in the Supplemental Materials. Notably, as shown in the Supplemental Materials, we observed a significant difference between trauma conditions on the first day, $F(1, 157) = 4.36$, $p = .038$, $\eta_p^2 = 0.03$, with participants in the man-made war condition reporting significantly higher vividness compared to those in the natural disaster condition, $t(161) = 2.09$, $p = .038$, $d = 0.33$. This finding suggests that different types of traumatic films may elicit varying intrusion-related vividness. Given the potential influence of film characteristics (such as the vividness of the film imagery) on our results, we therefore conducted mediation analyses separately within each trauma condition (natural disaster and man-made war) to more clearly examine the effects of attachment priming on daily outcomes.

Hypothesis 2: Multilevel Mediation Model in Daily Intrusion Count, Intrusion-Related Vividness, Intrusion-Related Distress, and Mediation Model in IES-R

We conducted between-person correlations among the main study variables, and the results are presented in Supplemental Table S4. Furthermore, we used multilevel mediation analyses to test whether T1 state attachment security and T2 reappraisal mediated the effects of attachment priming on daily intrusion count, vividness, and distress under both natural disaster and man-made war conditions.

The results showed that for daily intrusion count (see Figure 4 and Supplemental Tables S5–S6), there were no significant mediation effects in either the natural disaster or man-made war conditions. However, path analysis revealed that interpersonal attachment directly and negatively predicted daily intrusion count under the natural disaster condition. For daily intrusion-related vividness and distress under the natural disaster condition (see Figure 4A and

Figure 3
Estimated Means of Intrusive Symptoms



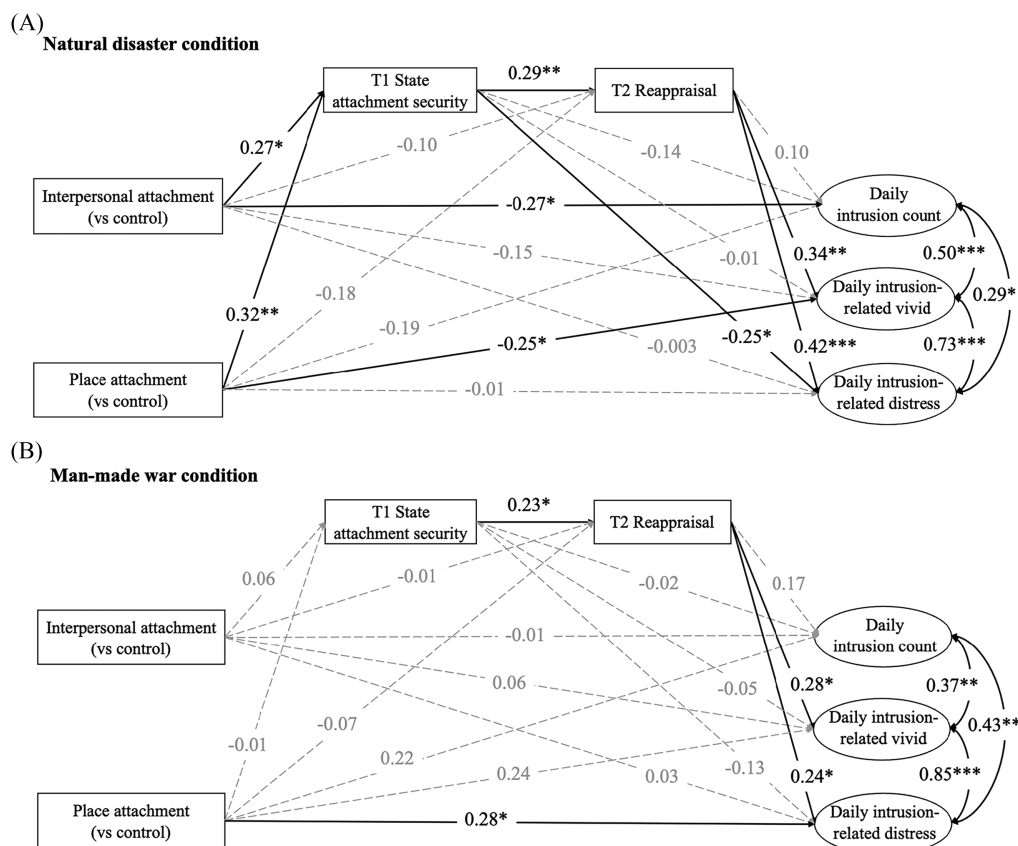
Note. Estimated means of (A) daily intrusion count, (B) daily intrusion-related vividness, and (C) daily intrusion-related distress across attachment priming conditions and trauma types at the within-person level. Interpersonal = interpersonal attachment security priming condition; place = place attachment security priming condition; control = control condition.

* $p < .05$. ** $p < .01$.

Supplemental Table S5), both interpersonal and place attachment priming (vs. control) had significant indirect effects on vividness (indirect effect = 0.08, $p = .025$, 95% CI [<0.001 , 0.27]; indirect effect = 0.10, $p = .013$, 95% CI [0.01, 0.31]) and distress (indirect

effect = 0.08, $p = .023$, 95% CI [0.001, 0.26]; indirect effect = 0.10, $p = .011$, 95% CI [0.01, 0.30]) through T1 state attachment security followed by T2 reappraisal. In addition, under the natural disaster condition, place attachment priming directly and negatively

Figure 4
Multilevel Mediation Models



Note. Multilevel mediation model examining the effects of attachment priming on daily intrusion count, vividness, and distress under (A) natural disaster and (B) man-made war conditions. Model shows standardized coefficients. Dotted lines represent nonsignificant paths, solid lines represent significant paths. T1 = Time 1; T2 = Time 2.

* $p < .05$. ** $p < .01$. *** $p < .001$.

predicted intrusion-related vividness. Although no significant mediation effects were found in the man-made war condition (see Figure 4B and Supplemental Table S6), place attachment priming directly and positively predicted intrusion-related distress.

Furthermore, mediation models were constructed to test whether T1 state attachment security and T2 reappraisal mediated the relationships between attachment priming and both IES and its intrusion subscale under both natural disaster and man-made war conditions (see Supplemental Figure S5 and Supplemental Table S7). The results showed that, under the natural disaster condition, interpersonal/place attachment negatively predicted IES-R (indirect effect = -2.30 , $p = .016$, 95% CI [-5.60 , -0.18]; indirect effect = -2.84 , $p = .006$, 95% CI [-6.38 , -0.54]) and IES-R intrusion (indirect effect = -0.60 , $p = .018$, 95% CI [-1.55 , -0.03]; indirect effect = -0.75 , $p = .007$, 95% CI [-1.77 , -0.11]) through increased T1 state attachment security. However, the direct effect of attachment priming on the IES-R and its intrusion subscale under the natural disaster condition was not significant, which may be attributed to a suppression effect (MacKinnon, 2008; Wen & Ye, 2014). In addition, no significant mediation pathways were observed under the man-made war condition.

Discussion

This study preliminarily explores the effects and psychological mechanisms of interpersonal and place attachment security priming on traumatic symptoms, with a specific focus on intrusive symptoms, under both man-made and natural traumatic conditions. After confirming the effectiveness of the manipulation, the findings indicated that the effects of activated interpersonal or place attachment on intrusive symptoms varied by trauma type and exhibited both immediate and short-term effects. While neither type of attachment security priming immediately alleviated distress caused by viewing the traumatic film, both exerted mitigating effects during the following week: Interpersonal attachment security reduced daily intrusion counts, and place attachment security lowered both intrusion counts and vividness, particularly in the context of natural disasters. However, the interaction between man-made war and place attachment security priming is associated with greater daily intrusion-related distress. Meanwhile, state attachment security and reappraisal mediated the effects of both types of attachment priming on the vividness and distress of daily intrusion-related memories, as well as traumatic symptoms 1 week later under natural disaster conditions. These findings address previous research gaps and preliminarily reveal the dual roles of interpersonal and place attachment security in intrusive memories, thereby offering potential possibilities for clinical interventions targeting intrusive and traumatic symptoms.

The Effects of Attachment Priming on State Attachment Security and Traumatic Symptoms

Although we found a nonsignificant main effect of interpersonal and place attachment security priming on immediate distress after viewing the trauma film (with a very small effect size), multilevel analyses indicated a significant main effect on daily intrusive memories during the subsequent week. This result indicates that

interpersonal and place attachment security priming has the potential to alleviate intrusive memories in the short term, consistent with previous studies (Bryant & Datta, 2019; Karl et al., 2021). Activating mental representations of interpersonal attachment security enhances perceived support, trust in others, and positive self-referential processing (Cassidy & Shaver, 2016). Similarly, activating place attachment fosters a sense of survival and security while enhancing belongingness (Bonaiuto et al., 2016; Giuliani, 2003; Scannell & Gifford, 2010). These effects can persist for several days and contribute to a reduction in the frequency of intrusive memories following the traumatic stimulus (Bryant, 2023).

However, we also found that state attachment security decreased after the trauma film in the attachment priming condition and was lower in the place attachment condition compared to the control, with medium effect sizes, which is contrary to expectations. This suggests that activation of attachment security did not confer an immediate protective effect on state attachment security following simulated trauma. The conservation of resources model (Hobfoll, 1989) offers a possible explanation for this phenomenon, proposing that individuals strive to retain, protect, and build valuable resources and perceive the loss of these resources as a significant threat. Priming attachment security may increase individuals' awareness of their emotional and social resources and may lead them to engage in efforts to protect them. In the context of trauma, this heightened awareness may intensify the perceived threat of losing such resources, thereby reducing state attachment security (Hobfoll, 1989; Solomon et al., 2008). Another possible explanation is that attachment security priming enhances affective empathy and compassion (Gillath et al., 2022), increasing emotional openness and reducing psychological defenses, which may render individuals more vulnerable to traumatic stimuli and temporarily decrease state attachment security. Although this finding is inconsistent with expectations, it remains plausible and further indicates that the mechanisms underlying interpersonal and place attachment security in the context of trauma are more complex and multifaceted than previously assumed, potentially encompassing both protective and vulnerability-enhancing effects.

Furthermore, compared to natural disaster films, viewing man-made war films results in a greater decrease and lower levels of state attachment security, while simultaneously leading to fewer daily intrusions in the following week. Man-made war scenarios often evoke a sense of responsibility and trigger complex, painful feelings such as anger, blame, and distrust (Liu et al., 2017; Zagefka et al., 2010), which may account for the marked reduction in state attachment security. This may facilitate immediate dissociative or cognitive defensive responses (e.g., detachment and emotional numbing), thereby reducing emotional processing and sensory encoding of the traumatic event and resulting in fewer intrusive memories (Brewin et al., 2010).

The key finding indicated that the interaction between activating interpersonal or place attachment and trauma type significantly predicted daily intrusive symptoms. Further simple effects analyses revealed that, among participants exposed to natural disaster films, daily intrusions were higher in the control condition than in those exposed to man-made war films. Within the natural disaster condition, activating either interpersonal or place attachment reduced intrusion counts over the following week compared with the control

condition. This result partially supports Hypothesis 1 and further validates the applicability of attachment theory (Bowlby, 1961, 1969, 1973) in natural disaster contexts. During such traumatic events, attached people and places provide a secure base and haven (Bonaiuto et al., 2016; Bowlby, 1969), promote connections among individuals as well as between individuals and places, foster a sense of belonging (Cushing et al., 2024; Scannell & Gifford, 2017a), thereby leading to fewer intrusions. These findings are consistent with previous research (Bryant & Chan, 2017; Tian et al., 2020).

Yet, for intrusion-related vividness and distress, activating place attachment security interacted with trauma type to significantly predict both outcomes. Participants who activated place attachment, compared with those in the control condition, reported fewer daily intrusion-related vivid images following exposure to natural disaster videos but greater daily intrusion-related distress after viewing man-made war films, which was contrary to Hypothesis 1. Recalling one's home or familiar places after a natural disaster may enhance feelings of connection, social cohesion, and cooperation (Silver & Grek-Martin, 2015), providing emotional buffering that reduces the impact of vivid intrusive imagery. In contrast, in the context of man-made war trauma, one possible explanation is that place attachment may enhance individuals' perceived control over their environment (Scannell & Gifford, 2017b), leading to deeper engagement that amplifies the differential effects between man-made and natural trauma. Participants exposed to man-made trauma are more likely to experience complex negative emotions (Liu et al., 2017; Zagefka et al., 2010), which may prompt painful memories associated with attached places, such as the loss of friends or changes in appearance and ownership (Scannell & Gifford, 2017b). These factors can further intensify fear regarding the loss of valuable resources (e.g., significant others and places; Hobfoll, 1989), thereby leading to higher intrusion-related distress. As an old saying goes, "The objects remain, but the people have changed; everything comes to a halt. Tears have already fallen when I want to express my grievances." This illustrates that place attachment carries with it a "false" sense of security (Bonaiuto et al., 2016) and has its own "shadow side" (Chawla, 1992).

State Attachment Security and Reappraisal as Mediators in the Relationship Between Attachment Priming and Traumatic Symptoms Under Man-Made and Natural Trauma Conditions

These findings suggest that interpersonal and place attachment security priming have similar effects on intrusive symptoms/PTSS, which may be mediated by state attachment security and reappraisal, thus partially supporting Hypothesis 2. To be specific, no significant mediation effect was found in the man-made war condition. Yet, after viewing natural disaster films, both interpersonal and place attachment priming demonstrated a mediating effect in reducing PTSS and its intrusion dimension in the subsequent week via postpriming state attachment security. This is different from our immediate results, which showed that neither type of attachment priming was able to provide an immediate mitigating effect on distress following traumatic stimuli. These contrasting findings may suggest that the effects of attachment security priming on trauma symptoms are delayed rather than immediate. This may be because, when faced with a sudden enhanced sense of threat caused by trauma

films, individuals may resort to dysfunctional coping strategies to maintain their own functioning (Dibbets, 2020; Ehlers & Clark, 2000), thereby preventing the immediate alleviation of distress. However, with the passage of time, the secure base and psychological benefits provided by the activation of state attachment security during writing may continue to exert their effects (Cushing et al., 2024; Scannell & Gifford, 2017b). As a result, people gradually adapt to and process traumatic situations, actively engage in positive thinking to discover meaning in the trauma, and then seek opportunities to transform threatening situations into favorable conditions (Cann et al., 2011; Sears et al., 2003), or seek help and support from their attachment figures (Mikulincer & Shaver, 2020). These strategies can help promote beneficial psychological changes after trauma, finally relieving trauma symptoms.

Meanwhile, both interpersonal and place attachment priming may increase the daily vividness and distress of intrusion-related symptoms by enhancing postpriming state attachment security and posttrauma reappraisal. Distinct from Hypothesis 2, we found that increased use of reappraisal following exposure to natural disaster films was associated with worsened daily intrusion-related vividness and distress. This finding contrasts with previous research indicating that the long-term use of reappraisal to alter perspectives effectively alleviates trauma (e.g., Sears et al., 2003; Woud et al., 2012). This discrepancy may be attributable to the timing and context of reappraisal. First, the use of reappraisal strategies in our study was measured immediately after simulated trauma activation, which reflects conscious efforts to modify participants' understanding of the traumatic events and their significance (Gross, 2014). This heightened conscious processing and encoding of trauma cues may intensify trauma memories in the short term and result in more vivid and distressing daily intrusive memories (Brewin et al., 2010). Furthermore, when reappraisal involves fundamental challenges to core beliefs about the self or world, such as reevaluating one's sense of belonging or worldview security, it may provoke profound distress and hopelessness (Hobfoll, 1989) in the short term, thereby increasing the likelihood of intrusive symptoms. Hence, we cautiously propose a possible interpretation for this phenomenon that painful and distressing memories may need to be revisited before they can be effectively processed and managed (Banich et al., 2009). Actually, this perspective is consistent with clinical observations from several effective trauma treatments, such as prolonged exposure and eye movement desensitization and reprocessing (Ford, 2018). In these approaches, people typically undergo a process of reappraisal early in treatment, which may involve shifting from passive immersion in re-experiencing inescapable traumatic events to active and intentional reflection on the trauma (Ford, 2018). This process requires people to revisit the traumatic scene, which elicits involuntary traumatic images and encourages them to systematically direct their attention to these memories (Brewin et al., 2010). Nevertheless, generalization of our findings requires great caution. Future interventions should carefully consider the timing and context of reappraisal and avoid implementing it immediately after trauma. For trauma survivors, reappraisal is best carried out under the supervision of professional mental health personnel, thereby minimizing the risk of exacerbating short-term intrusive symptoms.

It is noteworthy that interpersonal and place attachment priming were also found to differentially affect aspects of intrusive symptoms. After participants viewed natural disaster films, interpersonal

attachment priming was directly associated with a reduction in daily intrusion count. Meanwhile, place attachment priming was more closely related to the vividness and distress of intrusive memories. This may be because, in the context of natural disasters, interpersonal attachment can serve as a preventive mechanism, prompting active social connections (Xu, 2018) and fostering feelings of support and trust (Cassidy & Shaver, 2016; Charuvastra & Cloitre, 2008), thereby preventing the occurrence of intrusive memories. In comparison, place attachment security is more likely to serve as a coping mechanism after the occurrence of intrusive memories. Emotional bonds with familiar environments can fulfill psychological needs and promote positive emotions, providing comfort and a sense of security (Scannell & Gifford, 2017a, 2017b). This may buffer the emotional and sensory impact of intrusive memories, thereby reducing their vividness and distress.

Limitations and Implications

Our study has some limitations that require attention in future research. First, this study used nonclinical samples, which may introduce bias and warrant caution when generalizing to clinical populations, particularly those with trauma experiences. However, some lab-based analogue findings based on nonclinical people have shown promising utility and were applied to intervention development (Varma et al., 2024). Our findings may also help advance the development of trauma interventions based on secure attachment, which need future research to validate in clinical and trauma-exposed populations. Second, our study used only one traumatic film per condition, which may be a limitation. This is because differences observed between the man-made war and natural disaster conditions may not solely reflect trauma type but could also be influenced by specific characteristics of each film. For example, we found that participants in the man-made war condition reported significantly higher vividness than those in the natural disaster condition on Day 1 after viewing the films. Such differences in vividness could act as confounding factors and should be controlled for in future research designs. Thus, future studies should include multiple films per condition to ensure that observed effects are primarily attributable to trauma type rather than unique features of each film. Yet, it is important to note that exposing participants to multiple traumatic films per condition warrants particular ethical consideration, as this procedure may impose excessive cognitive and emotional burdens and potentially influence subsequent reports of intrusive memories (Brewin et al., 2010; Nixon et al., 2007). Third, our study focused on the effect of attachment security activation on intrusive symptoms, overlooking its potential impact on other PTSD symptoms, such as avoidance and hyperarousal, which warrants further investigation. Fourth, there are some confounders that may influence results but were not measured in this study, such as dissociation, which should be measured and controlled in future studies. Fifth, we only examined the immediate and short-term effects of attachment security priming. Future research should adopt a longitudinal perspective to investigate the short-term and long-term effects of interpersonal and place attachment security on intrusive symptoms and trauma recovery in the context of man-made or natural trauma. Sixth, our multilevel model did not incorporate days as a variable, which may limit our ability to capture temporal patterns in intrusion symptoms. Future studies should

consider including daily-level factors to more comprehensively examine temporal variations in intrusion symptoms.

Despite the above limitations, the present study holds important theoretical and practical implications. Regarding its theoretical contribution, this study provides preliminary evidence for the effects of activating interpersonal and place attachment security on post-traumatic responses to man-made war and natural disaster. Activation of both interpersonal and place attachment was associated with decreased daily intrusion counts. In the context of natural disasters, state attachment security and reappraisal mediated the impact of interpersonal and place attachment priming on traumatic symptoms, especially intrusive symptoms. In contrast, the interaction between man-made war and place attachment activation was linked to heightened intrusion-related distress. These findings also provide useful information about the immediate and short-term effects of attachment activation. Overall, these empirical findings extend the applicability of attachment theory (Bowlby, 1980) and conservation of resources model (Hobfoll, 1989) to the field of trauma psychology, shedding light on the possible role of the attachment system in relieving analogue trauma. They provide preliminary insights to facilitate further investigation into the impact of the attachment system on the prediction, development, and maintenance of PTSD.

For clinical implications, our findings provide empirical support for developing psychological interventions tailored to specific trauma types. When dealing with natural disasters, interventions that primarily focus on enhancing interpersonal attachment security, such as using text messages (Otway et al., 2014) or videos (Eguren et al., 2023), may effectively reduce intrusive symptoms. In contrast, man-made war should be approached with greater caution, particularly by avoiding the activation of place attachment, which may provide victims with a “false” sense of security (Bonaiuto et al., 2016) or reveal a “shadow side” (Chawla, 1992), thereby potentially exacerbating harm. Furthermore, caution is warranted when conducting immediate posttrauma reappraisal, as this process may increase trauma exposure and intrusive symptoms; thus, it should be implemented under the guidance of professional mental health practitioners.

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